Reading Shallow Flood Depth from Real USGS Gauge-Camera Footage with a Learned Staff-Plate Reader

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Abstract

How accurately can a fixed camera estimate flood water depth, and how well does that transfer across cameras? I use real USGS staff-plate cameras whose official gauge heights serve as labels. A learned reader registers each frame, rectifies the plate, classifies plate rows as submerged and decodes the water level; an image-quality gate abstains on bad frames. On the main shallow-flow camera (2259 frames, 15 months), leaving out whole months gives 1.13 cm mean absolute error on the 67% of frames passing the gate (6.97 cm for a colour-threshold baseline) and 0.79 cm on two locked months; error stays below twice native up to 5.0 cm per pixel. Applied unchanged to three cameras downloaded afterwards, with each camera held out of training in turn, it beats a no-water guess on two cameras and fails on one; a river camera is read to 12 cm over a 211 cm flood.

1 Problem and Motivation

Question. Given one frame from a fixed camera that sees a reference object of known scale, estimate the water depth h (cm) at the object. Any flood decision (an alarm threshold, severity classes) is a function of h, so I evaluate h itself and report error by depth range rather than fixing a threshold. One row is one frame at one camera and time. The prediction uses the current frame, a one-time setup (reference views, plate geometry) and, for pose recovery only, earlier frames; the gauge is never available at prediction time. Camera water-level work mostly targets rivers or deep urban water (Vandaele et al., 2021; Liang et al., 2023). The main camera (USGS 411229084541101, overland flow near Harlan, Indiana) sees shallow flow; other cameras test transfer (Sec. 7).

2 Data and Preparation

Source. Frames come from the public USGS National Imagery Management System bucket (USGS & repos, 2026); labels are the USGS gage height, both public domain. NWIS was not reachable from my environment, so the gauge values come from the committed result tables of two open-source projects (USGS & repos, 2026) (not re-checked against NWIS); no camera-derived value is a label. The plate spans 0–62 cm; the gauge sits beside it.

Content. 2259 frames (median spacing 21 min), 15 month blocks; gauge -0.6 to 55 cm; 35% dry, 25% infrared night; the pose changes between months.

Quality checks. SIFT registration (Lowe, 2004) to a reference view succeeds on 82% of frames (the rest reuse the last good pose); images show fog, mud or droplets, glare and over-exposure; the plate bottom is at 4.3 cm, so shallower water is unobservable.

Other cameras. After the method was frozen I added three staff-plate cameras (USGS 405327084555402, 411607084241200, 450157090060701; ≈1400 frames each, Jul–Nov 2026, gauge from the NWIS API), used only for evaluation, and a river camera (USGS 01421500, Hancock NY, portrait view) whose stage reaches the plate only in one flood (11.2–18.1 ft, 39 gated frames over 211 cm).

Preprocessing and leakage control. Frames are resized to 1280 px; the plate quad is rectified to a 360×48 strip (row ≈ 0.16 cm). Everything learned is fitted on training months only: the dry template (per-row median Lab colour of dry daytime frames), the affine calibration, the classifier, the baseline threshold and the gate thresholds (training Laplacian-variance quantiles). Registration reference views come from development months and use geometry, never labels.

3 Methods

Baselines. B0: always 0 cm. B1: a row is wet if its colour change from the dry template exceeds τ (tuned on training months), then the decoding below.

Learned reader. For frame i and strip row k (height h_k), a gradient-boosted classifier (scikit-learn `HistGradientBoosting`, Pedregosa et al., 2011) outputs $p_{ik} = P(\text{row submerged})$ from row colour, change from the template, texture, vertical context and frame quality. Labels are $y_{ik} = \mathbb{1}[h_k < g_i]$ (g_i the gauge height): no manual annotation. Water level is the monotone change point $b^\star = \arg\max_b \sum_{k<b} \log(1-p_{ik}) + \sum_{k\geq b} \log p_{ik}$, mapped to height and then to centimetres by a least-squares affine fit; levels at or below the plate bottom report 0. The log-likelihood gap to the runner-up boundary is the confidence *margin*.

Cross-camera protocol. For each target camera the classifier is trained on the *other* cameras only (labels from their nominal plate geometry); the target's dry template and gate thresholds come from its earliest month. *Zero-shot* reads depth with the nominal geometry; *calibrated* fits an affine map on that month; all other months are evaluated. For Hancock, calibration is one level offset from 30 frames just above the plate bottom, evaluated on the locked flood; *pooled* retrains with its development frames added.

Quality gate. A frame is rejected when its Laplacian variance lies outside $[0.55, 1.7]$ times the training median (low: fog, blur; high: lens dirt, glare).

4 Evaluation Design

The split unit is the calendar-month block, never the frame: consecutive frames are near duplicates and lighting and pose are shared within a month. Two blocks (2025-04, 2026-04) were locked in `make_usgs_splits.py` before any model was trained; the other 13 (1976 frames) support leave-one-block-out cross-validation (CV) and all tuning (classifier hyperparameters were left at fixed defaults). **Disclosure:** I looked at the locked blocks twice: the first gate was one-sided, dev errors then showed dirty-lens frames also fail, so the gate became two-sided and the run was repeated (both logged; the first gave 1.13 cm at 86% coverage). It is not a pristine single look.

The Hancock flood is locked and used for no template, offset or training. **Metrics.** Mean absolute error (MAE, primary), RMSE and bias (cm), overall and by gauge depth range (dry ≤ 0.5 cm, 0.5–10 cm, $>$10 cm). Uncertainty is the spread over held-out blocks (95% intervals bootstrap whole blocks; two locked blocks give none, so locked rows show point values).

5 Results and Error Analysis

Main result. Table 1 compares the reader with the baselines (always-0 cm: 8.80 cm CV, 18.90 cm locked). On gated frames in CV the learned reader has 1.13 cm MAE (95% CI 0.67–1.69; RMSE 3.1, bias +0.4 cm) versus 6.97 cm for the threshold baseline. On the two locked months it reaches 0.79 cm (baseline 11.52). Without the gate the errors are 3.38 cm (CV) and 3.11 cm (locked): the gate buys reliability by abstaining on 33% (CV) and 18% (locked) of frames. Per-month MAE has median 1.06 cm and worst 3.78 cm. (Smoothing hurts: a causal five-reading median raises MAE from 1.13 to 1.77 cm, since the level moves tens of centimetres per hour; headline numbers are per-frame.)

Table 1: Held-out depth error (cm). Gated = frames passing the quality gate (coverage 67% CV, 82% locked); last three columns: MAE by gauge depth range; CI: block bootstrap.

Eval.	Frames	Method	n	MAE [95% CI]	RMSE	bias	dry	0.5–10	>10
CV	gated	learned reader	1319	1.13 [0.7, 1.7]	3.1	+0.4	0.4	2.0	1.5
	gated	threshold baseline	1319	6.97 [6.0, 8.4]	10.1	+1.1	5.9	8.0	7.8
	all	learned reader	1976	3.38 [2.1, 4.5]	7.3	+0.5	1.9	5.0	4.1
Locked	gated	learned reader	233	0.79	1.2	+0.0	0.0	1.6	0.6
	gated	threshold baseline	233	11.52	14.6	-8.7	9.2	7.2	12.5
	all	learned reader	283	3.11	8.9	-2.1	1.6	2.1	3.4

Table 2: Gated MAE (cm) per camera; max = largest gauge (cm). In-site: leave-one-month-out. Held-out: classifier trained on the other cameras. "/ 0": the no-water guess on the same frames. † Hancock: range of the locked flood above the plate bottom; its guess is the plate bottom.

Camera	Scene	max	n	in-site / 0	held-out zero-shot / 0	held-out calib.
in_overland	cream plate	55	1319	1.13 / 8.80	6.69 / 11.16	7.14
in_north	cream plate	25	948	1.05 / 1.44	1.08 / 1.50	1.08
oh_wetland	white plate	42	1373	0.54 / 2.73	1.52 / 2.01	1.81
wi_mt1	orange plate	19	1342	0.55 / 0.27	4.96 / 0.33	4.96
hancock	river plate	211†	–	–	20 / 86	12

Other cameras (Table 2). Trained in-site by leave-one-month-out, the reader beats always-0 on in_overland, in_north (1.05 vs. 1.44 cm) and oh_wetland (0.54 vs. 2.73 cm); wi_mt1 stays almost dry (gauge $\leq$19 cm) and always-0 wins (0.27 vs. 0.55 cm). With the camera held out of training, zero-shot reading beats always-0 on in_overland (6.69 vs. 11.16 cm, but only 47% pass the gate), in_north (1.08 vs. 1.50) and oh_wetland (1.52 vs. 2.01) and *fails* on wi_mt1 (4.96 vs. 0.33 cm): never having seen an orange plate, the classifier reads it as fully submerged in sunny frames (69 cm at gauge 0). One month of calibration does not help. On the river plate (Fig. 1c) zero-shot MAE over the 211 cm flood is 20 cm (guessing the plate bottom: 86), 12 cm calibrated; pooling helps inside the stage range it saw (4, 7 vs. 8, 11 cm at 11–13, 13–16 ft) and hurts beyond it (37 vs. 22 cm at 16–19 ft).

Failure audit (Fig. 2, bottom). Infrared is *not* worse than daylight (3.30 vs. 3.41 cm); costly cases are a borrowed pose (9.82 vs. 1.91 cm; the camera moved) and frames outside the gate (7.90 vs. 1.13 cm). In the worst accepted day frame glare washes out the plate and water streaks on the panel look like wet rows.

6 Graduate Depth Analysis

Resolution sweep and error model. To vary the scale s, registration is computed once at native resolution (camera set up once), then each frame is area-downsampled by $f \in \{1, \ldots, 64\}$ and the plate re-sampled with $360/f$ rows, giving $s = 0.16f$ cm/px. The reader is retrained per f. Fig. 1d: gated MAE is 1.13 cm at the native scale, still 1.36 cm at 2.5 cm/px and 1.84 cm at 5.0 cm/px, but 3.61 cm at 10.0 cm/px, more than twice native; the baseline goes from 6.97 to 11.55 cm. With plate scale s and waterline error σ_v (px), $\mathrm{Var}(\hat{h}) = s^2\sigma_v^2 + \sigma_g^2$, where σ_g is gauge–plate offset and calibration error. Fitting $\mathrm{RMSE}^2 = (\sigma_v s)^2 + \sigma_g^2$ to the gated sweep gives $\sigma_v = 0.55$ px and $\sigma_g = 3.0$ cm. The σ_g floor dominates at fine scales, so resolution is not the binding constraint at this camera; the plate bottom and image quality are. Ranking frames by decode margin instead of the Laplacian gate gives 2.28 cm at the gate's coverage (gate: 1.13 cm), so a cheap image statistic abstains better than the classifier's own confidence.

7 Limitations and Conclusion

Each camera gives an effective sample size of months, not frames: five cameras, three never used for design, and the river camera contributes one flood (39 frames). Labels are a gauge beside the plate, not tape measures. The plate begins at 4.3 cm, so shallower water is unobservable and reported as 0.

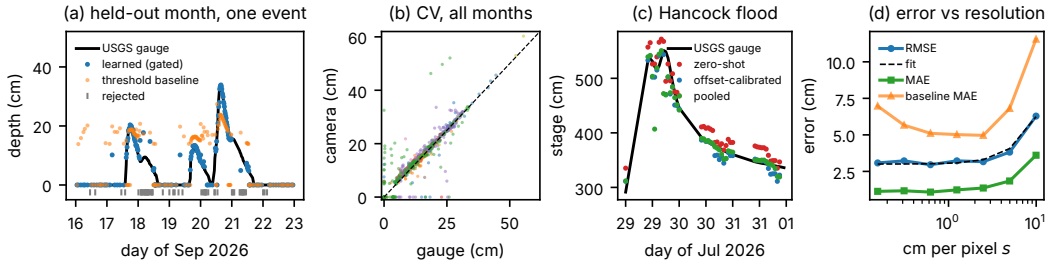


Figure 1: Real footage, held-out predictions. (a) One event in 2026-09 (model never saw this month); ticks: rejected frames. (b) Camera vs. gauge, gated frames, coloured by month. (c) River camera (Hancock), locked flood. (d) CV error vs. plate scale s; dashed: fitted error model.

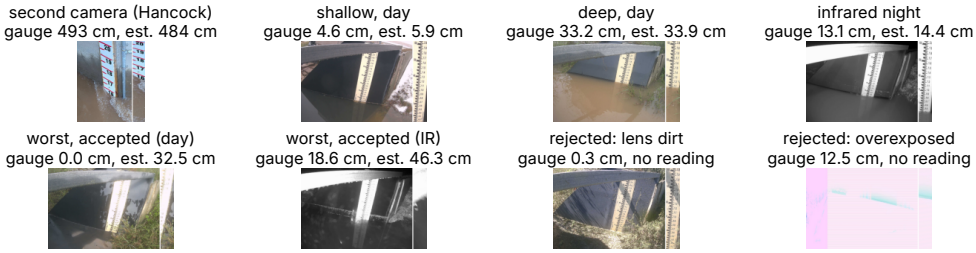


Figure 2: Real frames (crop and rectified plate). Top: median-error examples. Bottom: largest accepted errors and rejected frames (rules fixed in code).

The locked set is two months, seen twice, and the gate was designed with dev-month error in view. Stronger claims need more labelled cameras and tape-measured depths.

Conclusion. On real USGS footage a learned plate reader with a quality gate reads depth to about 1.13 cm (CV) on accepted frames, far better than a colour-threshold baseline, and stays within twice its native error up to about 5.0 cm/px; it abstains on 33% of frames and cannot observe depths below 4.3 cm.

8 Reproducibility and AI-Use Disclosure

Code, splits, gauge tables, the trained model and all results are in `github.com/kalpshah18/FloodCam` (branch `main`); `scripts/make_paper_assets.py` regenerates every number, table and figure here from `results/`. Frames are public domain; no credentials or private data are used. **AI use:** Claude Code (Anthropic) helped with brainstorming, code, debugging, experiments, the dashboard and drafting. I verified the pipeline with unit tests and by inspecting registration strips, failures and split leakage; I am responsible for all results.

References

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U.S. Geological Survey NIMS images and NWIS gage height (public domain); GitHub `waterline` (Apache-2.0), `stagesight` (MIT): gauge tables.

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